A First-Repair Mass Zero Without Angular-Moment Cancellation

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Abstract

We study two weighted histories in a specified outgoing-profile construction. For each fixed source Profile, we prove that every sufficiently small positive angular parameter admits a point strictly inside the first repair window where the mass history vanishes and the angular-weighted history is strictly positive at the same point. The positive parameter threshold may depend on the Profile. The decisive argument compares future moments of two separated repair bumps. At a mass zero, positivity of the second repair coefficient forces the first coefficient to be negative; the decreasing relative weight between the two histories then makes the angular tail negative and the accumulated angular history positive. We give the coefficient, boundary-sign, continuity and coordinate arguments needed to turn this conditional statement into an existence theorem. The corresponding source-bound propositions are formalized in Lean 4 and have a recorded fixed-checkpoint compilation. The result identifies a denominator obstruction for a prospective normalized ratio without requiring that ratio to be introduced into the source construction.

Keywords: weighted moments; localized repair; separated supports; formalized analysis; Lean 4.

1 Introduction

Moment repair can enforce several terminal cancellation conditions while leaving their partial histories markedly different. The location and sign of those partial histories matter when a derived quantity divides by one of them. A terminal identity alone does not determine whether cancellation occurs earlier, or whether a second history cancels at the same point.

This paper analyzes that question for the outgoing-profile construction defined in the pinned `NavierStokesAndEuler` source [1]. The two quantities are the source mass history M and its H-weighted history J. Our initial investigation considered the ratio J/(M H). The principal result is more naturally stated directly in terms of M and J: each fixed source Profile has a positive parameter neighborhood in which every positive parameter admits an interior first-repair point with M=0 and J>0.

The construction prescribes outgoing fields through localized pulse and repair terms. Its two terminal history constraints make partial cancellation a natural question: enforcing both histories at the end of the schedule need not synchronize them inside a repair window. Our analysis concerns that precise feature of the source construction. The names of the histories are used according to their defining integrals, without assigning an additional flux or vorticity interpretation.

The argument has two parts. First, moment comparisons determine the sign of the main-only second repair coefficient. The exact coefficient decomposition and continuity transfer this sign to the actual coefficient near zero. Boundary signs and the intermediate value theorem then produce an interior mass zero. Second, a relative-weight comparison proves J>0 at every first-window mass zero whose second coefficient is positive, not merely at a particular zero selected by the existence argument.

The mathematical contribution of this paper is the explicit source-linked derivation of this non-simultaneous cancellation and its formalization. The weighted comparison itself uses elementary real analysis. We do not claim priority over an exhaustive mathematical literature; the references establish the construction and formalization context, not a completed novelty survey. The proof and its precise quantifiers are the objects offered for assessment.

Section 2 specifies the construction and notation. Section 3 states the results. Sections 4 and 5 prove the conditional comparison and the existence theorem. Section 6 explains the formal correspondence and recorded computation. Section 7 discusses the interpretation of the result.

2 Construction and notation

2.1 Fixed source Profile

Fix a `ProfileF` of the source `NavierStokes.OutgoingProfile` namespace at revision

`8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538`.

A source Profile contains tail data, a coefficient bound and a reset witness. Its corrected amplitude is the source function `F.amp`; the coefficients contained in its angular-reset witness are not the axial repair coefficients studied below. All these objects belong to the same fixed Profile.

Write c for `F.data.core`. Its parameter structure includes P>0, m>0, 0<λ<1/10 and wait>25. We retain the full source Profile type, including its tail and reset data, rather than substituting an arbitrary amplitude satisfying only selected properties. Let

$$\begin{aligned} B = c.\mathrm{pulseStart}, \qquad L = \frac{13}{\lambda}, \qquad R = e^{L}, \\ a_0 = -\left(\frac{1}{2}+\lambda\right), \qquad a_1 = -\left(\frac{1}{2}+2\lambda\right) = a_0 - \lambda. \end{aligned} \tag{2.1}$$

The pulse-local radius is r; its source radial coordinate is $X = e^{B} r$. The source's logarithmic coordinate is y=B+log r.

The two repair intervals have endpoints

$$\begin{aligned} l_0 &= e^{L-63/20}, \quad u_0 = e^{L-57/20}, \\ l_1 &= e^{L-23/20}, \quad u_1 = e^{L-17/20}. \end{aligned} \tag{2.2}$$

The parameter bounds imply

$$1 < e^{11/\lambda} < l_0 < u_0 < l_1 < u_1 < R. \tag{2.3}$$

Indeed, L-63/20-11/λ=2/λ-63/20>0, and the remaining comparisons follow by comparing exponential arguments. In particular the first and second repair supports are strictly separated.

2.2 Source histories and pulse-local histories

The source defines the radial fields U_F, E_F and

$$H_F(X,\eta) = \sqrt{2X}\, E_F(X,\eta).$$

Its histories are

$$M_F(\eta, X) = \int_{(0,X]} U_F(u,\eta)\, du, \qquad J_F(\eta, X) = \int_{(0,X]} H_F(u,\eta) U_F(u,\eta)\, du. \tag{2.4}$$

These are the definitions of `Profile.M` and `Profile.J`, respectively [1]. J is an H-weighted history; no identification with vorticity or enstrophy is needed here.

For r≥1 define the local histories

$$\begin{aligned} \mathcal{M}_F(\eta, r) &= \text{massMoment}(c, F.\text{amp}, \eta, B + \log r), \\ \mathcal{J}_F(\eta, r) &= \text{angularMoment}(c, F.\text{amp}, \eta, B + \log r). \end{aligned}$$

The source identities `M_at_exp` and `J_at_exp` apply because B+log r≥0, and give the exact equalities

$$\mathcal{M}_F(\eta, r) = M_F(\eta, e^B r), \qquad \mathcal{J}_F(\eta, r) = J_F(\eta, e^B r). \tag{2.5}$$

There is no additional scale factor in (2.5). The integral prefactors introduced next belong to the histories themselves.

2.3 Repair coefficients and tails

Let $A_F(\eta)$=F.amp(η). It is smooth and strictly positive by the source corrected-amplitude results. Put s(η)=$(1+\eta^2)^{-1}$. Let

$$b_j(x) = \text{LocalizedMomentRepair.bump}(l_j, u_j, x), \qquad j = 0, 1.$$

Each b_j is nonnegative with topological support contained in (l_j,u_j). More precisely, the source bump is supported inside the middle half $[(3l_j + u_j)/4, (l_j + 3u_j)/4]$ of its declared window. Positivity throughout the larger window is not an assumption. Define c_j(η) to be component j of the source axial repair coefficient vector

$$\text{LocalizedMomentRepair.coefficients}\big(c.\text{exponents}, c.\text{lower}, c.\text{upper}, \text{debt}(c, A_F, \eta)\big).$$

The pulse is

$$p_F(\eta, x) = A_F(\eta)\, \text{mainPulse}(\lambda \log x) + c_0(\eta) b_0(x) + c_1(\eta) b_1(x). \tag{2.6}$$

For x≥l_0 the main pulse is zero. Thus every future interval starting inside the first repair window sees only the two correction bumps.

Let κ_i=`momentScaleci` and define the strictly positive prefactors

$$K_0(\eta) = \kappa_0 s(\eta), \qquad K_1(\eta) = \kappa_1 s(\eta)^2.$$

The source prefix constants P_M=`prefixMc` and P_J=`prefixJc` give, for 1≤r≤R,

$$\begin{aligned} \mathcal{M}_F(\eta, r) &= P_M \eta + K_0(\eta) \int_1^r x^{a_0} p_F(\eta, x)\, dx, \\ \mathcal{J}_F(\eta, r) &= P_J \eta s(\eta) + K_1(\eta) \int_1^r x^{a_1} p_F(\eta, x)\, dx. \end{aligned} \tag{2.7}$$

Terminal cancellation, supplied by `pulse_closes_prefix`, changes these to

$$\mathcal{M}_F(\eta, r) = -K_0(\eta) \int_r^R x^{a_0} p_F(\eta, x)\, dx, \qquad \mathcal{J}_F(\eta, r) = -K_1(\eta) \int_r^R x^{a_1} p_F(\eta, x)\, dx. \tag{2.8}$$

For r∈(l_0,u_0), abbreviate

$$T_j(r) = \int_r^R x^{a_0} b_j(x)\, dx, \qquad S_j(r) = \int_r^R x^{a_1} b_j(x)\, dx. \tag{2.9}$$

Then (2.8) becomes

$$\mathcal{M}_F = -K_0(c_0 T_0 + c_1 T_1), \qquad \mathcal{J}_F = -K_1(c_0 S_0 + c_1 S_1). \tag{2.10}$$

All terms in each equation refer to the same F, η and r.

3 Main results

Theorem 3.1 (conditional angular mismatch). Fix a source Profile F. For any real η and any r satisfying l_0<r<u_0,

$$\mathcal{M}_F(\eta, r) = 0 \quad \text{and} \quad c_1(\eta) > 0 \quad \Longrightarrow \quad \mathcal{J}_F(\eta, r) > 0.$$

No sign assumption on η is required in this theorem.

Theorem 3.2 (same-point existence). For every fixed source Profile F there exists ϵ_F>0 such that, for every 0<η<ϵ_F, there exists r∈(l_0,u_0) with

$$M_F(\eta, e^B r) = 0 \quad \text{and} \quad J_F(\eta, e^B r) > 0. \tag{3.1}$$

The positive threshold ϵ_F may depend on F, and the witness radius r may depend on η. Both clauses in (3.1) use that one witness.

4 Proof of the conditional comparison

Lemma 4.1 (tail signs and coefficient sign). At a point satisfying the hypotheses of Theorem 3.1, T_1>0, T_0>0 and c_0<0.

Proof. Since r<u_0<l_1 and u_1<R, the future interval contains the whole second support. Therefore T_1 equals its whole-line weighted bump moment. The source matrix-entry identity expresses this moment as a positive template row moment times a positive exponential. Hence T_1>0. Also T_0≥0 because its integrand is nonnegative on positive x. The mass-zero identity in (2.10), with K_0>0, gives

$$c_0 T_0 = -c_1 T_1 < 0. \tag{4.1}$$

If T_0=0, (4.1) is impossible. Thus T_0>0, and (4.1) forces c_0<0. □

Lemma 4.2 (comparison of relative weights). For every r∈(l_0,u_0),

$$u_0^{-\lambda} T_0 \le S_0, \qquad S_1 \le l_1^{-\lambda} T_1, \qquad l_1^{-\lambda} < u_0^{-\lambda}. \tag{4.2}$$

Proof. The identity a_1=a_0-λ gives $x^{a_1} = x^{a_0} x^{-\lambda}$ for x>0. The function x↦$x^{-\lambda}$ strictly decreases since λ>0. Wherever b_0 is nonzero, x<u_0 and thus $x^{-\lambda} \ge u_0^{-\lambda}$. Wherever b_1 is

nonzero, x>l_1 and $x^{-\lambda} \le l_1^{-\lambda}$. Multiplying these inequalities by the respective nonnegative integrands x^{a_0}b_j(x) and integrating gives the first two comparisons. The final strict comparison follows from $u_0<l_1$. □

Proof of Theorem 3.1. By Lemma 4.1, $c_0<0$ and $c_1>0$. Consequently the first inequality in (4.2) reverses when multiplied by c_0. Using both tail bounds and then (4.1),

$$\begin{aligned} c_0 S_0 + c_1 S_1 &\le c_0 u_0^{-\lambda} T_0 + c_1 l_1^{-\lambda} T_1 \\ &= c_1 T_1 (l_1^{-\lambda} - u_0^{-\lambda}) < 0. \end{aligned} \tag{4.3}$$

Equation (2.10) and $K_1>0$ now give $\mathcal{J}_F(\eta, r) > 0$. □

The argument explains why moment cancellation is not transported from one weight to the other: a negative first-support contribution receives a larger relative weight than the positive second-support contribution. Positivity of T_0 is a consequence of mass balance, not an additional assumption. It does not imply $b_0(r)>0$ at the zero itself.

5 Coefficient positivity and interior existence

5.1 Sign of the actual coefficient near zero

Write β_i=a_i+1 and q=L-3. The source main moments and their normalized values are

$$\mu_i = \int_1^R x^{a_i} \mathrm{mainPulse}(\lambda \log x)\, dx, \qquad \mathcal{F}_i = e^{-\beta_i q} \mu_i.$$

Let A_i=`rowMoment(c.exponentsi)` be the positive source template moments and k=$e^{-3\lambda/20}$.

The notation corresponds to `OutgoingPulseBounds.betaci`, `OutgoingPulseBounds.centerc0`, `OutgoingSchedule.mainMomentci`, `NSRW.P2.normalizedMainMomentci` and `OutgoingPulseBounds.rowMoment(c.exponentsi)`, respectively. Exact locators are the pinned definitions of beta, center and rowMoment, together with the NSRW normalized-moment definition at the execution revision. Thus q is the first repair center in logarithmic coordinates; it is not the endpoint l_0 or u_0. The formalized main-pulse and template comparisons give

$$\mathcal{F}_1 > 0, \qquad \mathcal{F}_0 < k\mathcal{F}_1, \qquad kA_1 \le A_0. \tag{5.1}$$

For completeness, the strict main comparison first bounds the extra exponential on the short logarithmic interval $0\le y\le 11/\lambda$ by $e^{-2+3\lambda}$. The condition $\lambda<1/10$ implies $-2+3\lambda<-3\lambda/20$. Multiplication by the positive normalized first moment provides strictness. The template comparison uses the nonnegative template support in $x>e^{-3/20}$ and the factor x^{λ}. These are separately formalized comparisons [2].

Multiplying (5.1) by the appropriate positive factors yields

$$\mathcal{F}_0 A_1 < k\mathcal{F}_1 A_1 \le \mathcal{F}_1 A_0.$$

Thus

$$\Delta = \frac{\mathcal{F}_0}{A_0} - \frac{\mathcal{F}_1}{A_1} < 0. \tag{5.2}$$

The normalized source repair equations identify the main-only second coefficient m_1 as

$$m_1 = \frac{-\Delta}{e^{2\beta_0} - e^{2\beta_1}} > 0. \tag{5.3}$$

The denominator is positive because β_0-β_1=λ>0. Explicitly, let

$$p_1 = \text{affineCoefficients}(c, 1, 0, 1),$$
$$m_1 = \text{affineCoefficients}(c, 0, 1, 1).$$

Both are constants determined by the fixed core c, independent of η. The actual coefficient of the fixed Profile satisfies the exact decomposition

$$c_1(\eta) = \eta(1 + \eta^2)p_1 + A_F(\eta)m_1. \tag{5.4}$$

In particular $c_1(0)$=$A_F(0)m_1$>0. The corrected amplitude is smooth, so (5.4) is continuous in η. The preimage of the positive half-line is an open neighborhood of zero. It therefore contains (-ϵ_F,ϵ_F) for some ϵ_F>0. Restricting to 0<η<ϵ_F supplies the required sign $c_1(\eta)$>0.

5.2 Boundary signs and continuity

Lemma 5.1 (first-window boundary signs). If η>0 and $c_1(\eta)$>0, then

$$\mathcal{M}_F(\eta, l_0) > 0, \qquad \mathcal{M}_F(\eta, u_0) < 0. \tag{5.5}$$

Proof. The source prefix constant satisfies

$$P_M = 4 + \int_0^B e^y \,\text{dropCoefficient}(c.m, y)\, dy > 0,$$

because the integrand is nonnegative. Both repair bumps vanish before and at l_0, so the partial pulse integral in (2.7) at l_0 is nonnegative: A_F>0, the main pulse is nonnegative for nonnegative argument, and x^{a_0}>0 on x≥1. The term $P_M\eta$ is strictly positive.

At u_0, the main pulse and first bump vanish on the future interval. The mass-tail identity reduces to

$$\mathcal{M}_F(\eta, u_0) = -K_0(\eta)c_1(\eta) \int_{u_0}^R x^{a_0} b_1(x)\, dx < 0,$$

since the second-bump moment is strictly positive. □

Lemma 5.2 (continuity on the repair interval). For fixed F and η, the map r↦$\mathcal{M}_F(\eta, r)$ is continuous on [l_0,u_0].

Proof. The source logarithmic mass primitive is

$$4\eta + \int_0^y e^t \,\text{axial}(c, A_F)(t, \eta)\, dt.$$

Its integrand is continuous, so it is continuous in y. The map r↦B+log r is continuous on [l_0,u_0], where r>0. Composition proves the assertion. □

Proof of Theorem 3.2. Choose ϵ_F from (5.4) and continuity. For every 0<η<ϵ_F, Lemmas 5.1 and 5.2 and the intermediate value theorem give a zero in [l_0,u_0]. The strict signs in (5.5) exclude both endpoints, so the zero belongs to (l_0,u_0). Theorem 3.1 gives positive local angular history at that very zero. Equation (2.5) transfers both statements to the same source radial coordinate $e^B r$. □

6 Formalization and computational record

The proofs use Lean 4 [3] with the source and dependencies fixed by the recorded environment. The theorem-to-declaration correspondence is given in Appendix A. The final declaration binds one F and one η, obtains a single radius from the conditional intermediate-value theorem, and passes that radius to the angular-positivity theorem. It does not combine separately chosen existential witnesses.

The recorded compilation at NSRW revision `3a50b5f50b034319a469b9de68f556be70ff f02c` used Lean 4.34.0-rc2 and the source revision of Section 2. The 12 October 2026 record comprises a source-target build and fourteen fresh external modules. All fifteen stages exited zero with empty stderr. Nineteen reported axiom surfaces for the final local, source-facing and composition declarations contain `propext`, `Classical.choice` and `Quot.sound`, with no `sorryAx` [4, 5]. A retained nonfatal stdout linter warning is documented in those records.

The formal compilation establishes the stated Lean propositions relative to the imported definitions and proof environment. The prose proof above explains their mathematical mechanism. The execution reused the pinned upstream cache; it was not a fresh rebuild of the entire upstream library. Execution and archive review were carried out within one operator trust domain. These facts describe the computational record, not an external peer-review process.

7 Interpretation and further questions

The result distinguishes terminal cancellation from intermediate behavior. M and J can both satisfy prescribed endpoint cancellation while having different signs or zeros inside a repair interval. The proof identifies the responsible structure: opposite-signed repair coefficients, separated supports and a strictly decreasing relative weight.

For a prospective real ratio Q=J/(M H), the point in Theorem 3.2 fails the condition M H$\neq$0 regardless of H at that point. This is a domain obstruction. It does not supply a quotient limit, divergence rate or singularity classification. In particular, total real division in Lean should not be confused with a partial-domain ratio used in analysis.

The present theorem is existential. It does not compute ϵ_F or r. Stronger zero geometry calls for a derivative argument and localization inside the actual positive region of the first bump, rather than positivity throughout its larger declared window. In particular, the positive integral T_0 alone does not establish positive bump density at the witness. These observations identify a concrete route toward investigating monotonicity, uniqueness and crossing behavior; those statements are not added to the present formal result. The full zero set outside the positive parameter neighborhood is another open question.

8 Conclusion

In the fixed outgoing-profile construction, an interior mass cancellation is accompanied by strictly positive angular-weighted history whenever the second repair coefficient is positive. Moment comparison, coefficient continuity and boundary signs ensure that such a point exists for every sufficiently small positive parameter of each fixed source Profile. The proof exposes a concrete mechanism for non-simultaneous cancellation and connects that mechanism to exact source-facing formal propositions.

Code availability and authorship

The formal input files and replay runner are identified by NSRW revision `3a50b5f50b034319a469b9de68f556be70fff02c` [2]. The two post-execution records [4, 5] are separate documents; their text is not asserted to exist at the execution revision. The accompanying reproduction guide identifies the selected inputs and receipt snapshots supplied with the local distribution candidate. Original raw execution evidence is privately retained. No published download or artifact DOI is claimed here.

Yun Kwansub is the named human author. AI assistance was used in analysis, formal-code development and manuscript preparation. The source construction and imported formal mathematics are attributed to their source repositories and authors.

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References

[1] OpenAI and repository contributors. *NavierStokesAndEuler*. Source revision `8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538`, especially `OutgoingSchedule.lean`, `LocalizedMomentRepair.lean`, `OutgoingPulseBounds.lean` and `OutgoingProfile.lean`. Fixed source: https://github.com/openai/NavierStokesAndEuler/tree/8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538.

[2] NSRW repository. *NavierStokes Research Workbench*: formal inputs and replay runner at revision `3a50b5f50b034319a469b9de68f556be70fff02c`. Fixed repository: https://github.com/flamehaven01/NavierStokes-Research-Workbench/tree/3a50b5f50b034319a469b9de68f556be70fff02c.

[3] Leonardo de Moura and Sebastian Ullrich. The Lean 4 Theorem Prover and Programming Language. In *Automated Deduction—CADE 28*, LNCS 12699, 2021. doi:10.1007/978-3-030-79876-5_37.

[4] NSRW. *P2 Z-core Colab execution receipt*, 12 October 2026. Local document (`docs/stage_receipts/P2_Z_CORE_COLAB_2026-10-12.md`).

[5] NSRW. *P2 Z+J Colab execution receipt*, 12 October 2026. Local document (`docs/stage_receipts/P2_ZPLUS_J_COLAB_2026-10-12.md`).

A Formal correspondence

Module filenames and local navigation are given below. All listed proof modules are also available under `formal/p2/` at the fixed execution revision [2].

Mathematical statement	**Lean declaration in `NSRW.P2`**	**Module**
Local mass partial/tail identities	`localMass_partial_identity`, `localMass_tail_identity`	`formal/p2/P2_Z01_LocalMassIdentities.lean`
Local angular partial/tail identities	`localAngularMoment_partial_identity`, `localAngularMoment_tail_identity`	`formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean`

Mathematical statement	Lean declaration in NSRW.P2	Module
Strict support separation	first_upper_lt_second_lower	formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean
Future mass balance	firstRepair_mass_zero_tail_balance	formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean
$T_0>0$ and $c_0<0$	firstBumpTail_pos_and_firstCoeff_neg	formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean
Relative-weight inequalities	firstBump_angularTail_lower, secondBump_angularTail_upper	formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean
Theorem 3.1	angularMoment_pos_of_firstRepair_mass_zero	formal/p2/P2_ZPLUS_J_FirstRepairAngularMoment.lean
Coefficient positivity near zero	actualSecondRepairCoefficient_pos_small_positive	formal/p2/P2_C04_SmallPositiveEta.lean
Strict boundary signs	localMass_lower_pos, localMass_upper_neg_of_secondCoeff_pos	formal/p2/P2_Z02_BoundaryMassSigns.lean
Conditional interior mass zero	exists_firstRepair_zero_of_secondCoeff_pos	formal/p2/P2_Z03_ConditionalInteriorZero.lean
Source mass coordinate bridge	localMass_eq_profileM	formal/p2/P2_Z04_FirstRepairInteriorZero.lean
Source angular coordinate bridge	localAngularMoment_eq_profileJ	formal/p2/P2_ZPLUS_J_SourceComposition.lean
Theorem 3.2	profileMJ_firstRepair_zero_small_positive	formal/p2/P2_ZPLUS_J_SourceComposition.lean

B Reproduction identity

Item	Recorded identity
NSRW execution revision	3a50b5f50b034319a469b9de68f556be70fff02c
Upstream revision	8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538
Dependency manifest Git blob	f07a8454cb6200d90bcc4371bc9965e9f8f46c7d
Lean	4.34.0-rc2
Replay runner	scripts/run-p2-z-replay.py
Reference execution	nsrw-z-phase-replay-20261012T065216884451Z

The runner requires the pinned source environment and exact NSRW checkout. It does not install dependencies. A new execution will have its own logs, timings and generated artifact hashes. Detailed invocation and distributed-evidence information is in the reproduction guide (research-papers/REPRODUCE.md) and the two records [4, 5].